

Electrical Connected Orchestra: A New Baton System that can Interactively Control the Body Movements of Performers

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Figure 1: Interactive Automatic Performance System based on the Proposed System

Abstract

This research leads to the creation of an innovative music performance system that connects conductors and performers. We developed an automatic performance system, that can interactively control the body movements of performers. This system converts music arranged by melody morphing into electrical muscle stimulations (EMS) and controls the body movements of multiple people with devices attached to their hands and feet to realize the performance. In addition, by controlling the behavior of the EMS using a baton interface with a built-in acceleration sensor, the conductor can interactively change the melody, tempo, and velocity of the performance.

CCS Concepts

• **Human-centered computing** → **Human-computer interaction (HCI)**.

Keywords

Human Augmentation; Interactive Musical performance System; Electrical muscle stimulation (EMS)

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1 Introduction

This paper describes an interactive music system, called “Electrical Connected Orchestra” which enables melodies to be played by performers who are manipulated by a conductor. In the system, the user controls the performers with electrical muscle stimulation (EMS) to perform the melody. In addition, by controlling the behavior of the EMS using a baton with a built-in acceleration sensor, the conductor can interactively change the melody, tempo, and velocity of the performance.

Variations of melodies are composed on the basis of the generative theory of tonal music (GTTM) [3], whereby switching the melody segments keeps the overall structure of the melody and only changes the ornamentation[1]. The time-span tree, which is the deep structure obtained as a result of a GTTM analysis, structurally shows the relationship between the important notes and the ornamentation notes. One example of a melody operation using the time-span tree is melody morphing, which generates an intermediate melody from two input melodies (Fig. 2) [2].

2 EMS System

The Electrical Connected Orchestra uses EMS to establish control over simple movements, such as snapping wrist joints and flexing elbow joints, of multiple people. The system includes a device that

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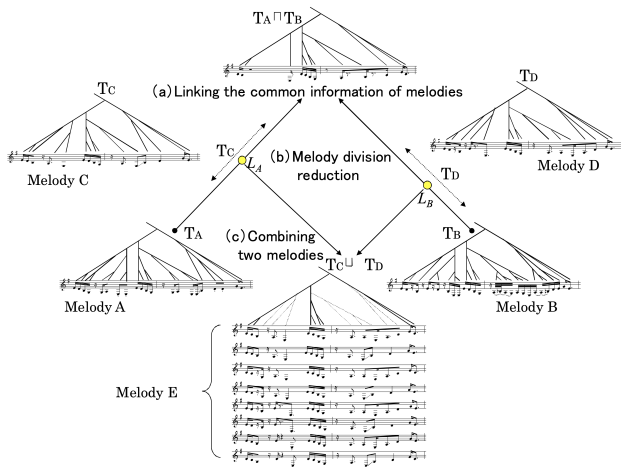


Figure 2: Melody Morphing Method.

generates electrical stimulation by using an Arduino, an electrical stimulation circuit, a PC, a DC stabilized power supply, and electrode pads, with reference to [4] (Fig. 3). In the circuit, a 555 timer is used to generate pulses, and a transformer is used to increase the voltage. The stimulation pulse frequency is 40 to 70 Hz and the pulse width is 0.8 ms. The pulse voltage can be controlled by the user and varied depending type of muscle to be exercised.

The system synchronizes the body movements of many people by controlling multiple EMS generators in an integrated manner. We evaluated the time response of the developed system. The average delay in transmission was 22.8 ms (standard deviation 6.4 ms) when stimuli were given to 10 subjects (8 males and 2 females) at the same timing. The subjects showed minor differences in their perception of the stimuli and magnitude of their movements, but these differences were eliminated by calibration.

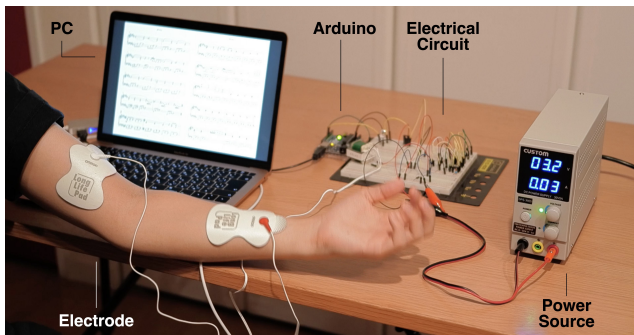


Figure 3: System Configuration of EMS System.

3 EMS Baton Interface

We designed a baton interface to control the EMS system, called EMS Baton. Fig. 4 shows the interface and its operation screen. The EMS baton has a built-in 6-axis acceleration sensor module (M5stick c plus) in the grip and acquires the movement of the conductor's arm swing. The tempo was measured by setting a threshold and

performing a discrete Fourier transform. The swing width of the baton corresponds to the velocity. We also designed a glove with a built-in acceleration sensor. By manipulating the position of the hand, this glove can change the next melody to be played. Thus, the conductor can interactively change the melody, tempo, and velocity of the performance by using the EMS Baton and glove.

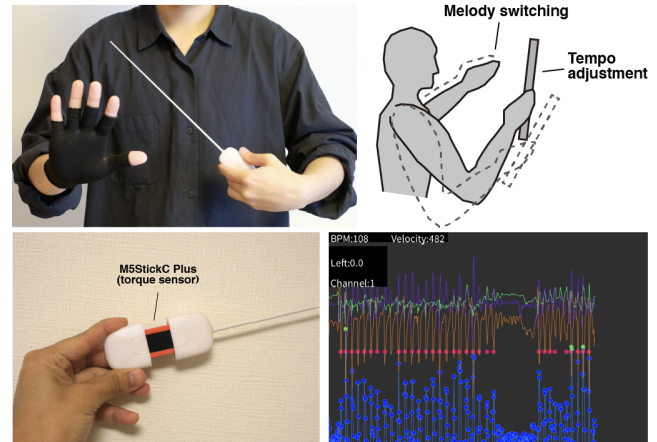


Figure 4: Interface and Operation Screen of EMS Baton.

Electrical Connected Orchestra is ideal for the automatic playing of ensemble percussion instruments (Fig. 1). We can convert music arranged by melody morphing into EMS signals to control the body movements of multiple people with devices attached to their hands to realize the performance. The melodies used in the system are divided into multiple segments, and each segment has multiple melody variations. The conductor can control performers by freely manipulating the melody, by changing the position of the gloved hand. Thus, the conductor can interactively change the melody, tempo, and velocity of the performance by using the EMS Baton and glove.

4 Conclusion

Electrical Connected Orchestra, which enables the user to manipulate music freely, uses a melody morphing method that we have constructed on the basis of melody manipulations using the time-span tree of GTTM. The conductor can interactively change the melody, tempo, and velocity of the performance. We will pursue a new style of musical experience by fusing music information processing and human physical functions. We plan to construct a system that enables more complex instrumental performances.

References

- [1] Masatoshi Hamanaka, Keiji Hirata, and Satoshi Tojo. 2007. Implementing 'A Generating Theory of Tonal Music'. *Journal of New Music Research* 35, 4 (2007), 249–277.
- [2] Masatoshi Hamanaka, Keiji Hirata, and Satoshi Tojo. 2008. Melody Morphing Method Based on GTTM. In *Proceedings of the 2008 International Computer Music Conference (ICMC'08)*. 89–92.
- [3] Fred Lerdahl and linguist Ray Jackendoff. 1983. *A Generative Theory of Tonal Music*. Cambridge, Massachusetts: MIT Press.
- [4] Hiroya Miura and Masatoshi Hamanaka. 2021. Deep Augmented Performers: A New Ensemble Performance System by Fusion of Melody Morphing and Body Movements. In *ACM UIST 2021 (UIST'21 Adjunct)*. ACM, 129–130.